

Enterprise Database Migration Architecture Guide

Cloud Infrastructure Operations - Standard Operating Procedure - Ref: ARCH-DB-MIG-703

1.1 Executive Architecture Overview

This technical standard defines the engineering methodology, infrastructure requirements, and execution runbook for migrating mission-critical relational databases (PostgreSQL and Oracle workloads) from legacy on-premises datacenters to cloud-native managed Aurora PostgreSQL. The objective is to achieve zero-loss, near-zero downtime cutover (less than 120 seconds of write suspension) while preserving data integrity, foreign key invariants, transactional consistency, and analytical audit compliance across multiple petabytes of operational data.

1.2 Pre-Migration Schema Assessment & SCT Workflow

Prior to data migration, database administrators must execute an automated static schema compatibility audit using the AWS Schema Conversion Tool (SCT). Key evaluation criteria include: stored procedure conversion from PL/SQL to PL/pgSQL, user-defined type conversions, partitioning schemes (range vs hash), and index conversion (converting proprietary bitmap indexes into B-Tree, BRIN, or GIN indexes). Custom PL/SQL packages with nested autonomous transactions must be refactored into microservice application logic or asynchronous queue workers.

1.3 Network Infrastructure and Bandwidth Sizing

A dedicated private network transit tunnel is mandatory. Teams must provision a dual redundant 10 Gbps Cloud Interconnect or AWS Direct Connect private virtual interface (VIF) between the on-premises core switch and the cloud Virtual Private Cloud (VPC). Network round-trip latency (RTT) must remain strictly under 8 milliseconds under peak load. Jumbo frame MTU (9000 bytes) must be enabled across all intermediate routers to maximize bulk TCP streaming throughput and minimize packet fragmentation.

1.4 Baseline Snapshot Export Utilities

The initial dataset snapshot is extracted using parallel `pg_dump` utilities configured with 16 parallel workers (`-j 16`) and directory format (`-F d`). Data streams are compressed on-the-fly using `zstandard` compression level 3 and uploaded directly to cloud object storage via multi-part upload pipelines. Secondary index creation and foreign key constraints must be disabled on the target cluster during bulk restore to achieve maximum write ingestion speeds.

Database Migration (CDC Streaming & Data Validation)

2.1 Change Data Capture (CDC) Architecture

To capture incremental database transactions while the baseline snapshot is being transferred and restored, engineers deploy a real-time CDC pipeline. The pipeline is powered by Debezium connectors running inside an Apache Kafka Connect cluster. On the source PostgreSQL engine, logical replication decoding is configured using the native 'pgoutput' plugin with a persistent replication slot named 'debezium_cdc_slot'. Source database parameters must set 'wal_level = logical', 'max_replication_slots = 10', and 'max_wal_senders = 10'. Debezium reads the PostgreSQL Write-Ahead Log (WAL) and publishes row-level change events (INSERT, UPDATE, DELETE) as serialized Avro records into Kafka topics.

2.2 Replication Lag Monitoring & Cutover Thresholds

A dedicated Kafka consumer sink deploys JDBC connectors writing to the target Aurora PostgreSQL cluster with exactly-once transactional semantics. Replication lag is measured continuously using Prometheus exporters tracking the difference between source current LSN (Log Sequence Number) and target applied LSN. Strict Pre-Cutover Condition: The CDC consumer replication lag must remain strictly under 250 milliseconds for at least two (2) continuous hours during normal business traffic before the change advisory board will grant authorization to initiate live application cutover.

2.3 Cryptographic Data Validation and Checksum Audit

Before executing final cutover, an automated audit verification harness performs dual validation: (1) Complete 100% row-count equality verification across all source and target tables, verified against information_schema metadata; (2) Randomized cryptographic row-hashing (MD5/SHA256 concatenation of primary key and payload columns) across 5% sample blocks of high-velocity transaction tables. Any checksum disparity exceeding 0.00% will halt cutover proceedings immediately.

Database Migration (Cutover Runbook & Rollback Safeguards)

3.1 Scheduled Cutover Execution Window

The production cutover window is scheduled strictly during the lowest traffic volume period: Sunday between 02:00 UTC and 04:00 UTC. All background cron jobs, scheduled analytics batch pipelines, and automated reporting workers must be paused 30 minutes prior to window initiation.

3.2 Step-by-Step Cutover Procedure

Phase 1 (T-48 Hours): Reduce DNS Time-To-Live (TTL) on database endpoint CNAME records to sixty (60) seconds to enable rapid client IP propagation. Phase 2 (T-00:00): Place source on-premises database in READ ONLY mode (`ALTER SYSTEM SET default_transaction_read_only = on`). Phase 3 (T+00:03): Wait for Debezium Kafka CDC pipeline to drain pending WAL mutations to zero (0 ms lag). Phase 4 (T+00:08): Re-enable target table foreign key constraints and secondary index validations. Phase 5 (T+00:15): Update application connection strings and DNS routing to point to target cloud Aurora cluster, and restore application write traffic.

3.3 Automated Rollback Triggers and Safeguards

An immediate, unconditional automated rollback to the source on-premises database is mandatorily triggered if any of the following occur: (a) CDC replication queue fails to drain to zero within 120 seconds of entering READ ONLY maintenance mode; (b) Post-cutover API Gateway HTTP 5xx error rate exceeds 0.05% for three (3) consecutive minutes during live canary verification; (c) Application database query latency spikes by more than 300% on p99 metrics or database deadlocks exceed 50 per minute. Reverse CDC Fail-safe: To guarantee zero-loss rollback, a reverse Debezium CDC pipeline streams all new writes from the target cloud database back to the on-premises database for seventy-two (72) hours post-cutover.